

Module 15 - chapter 7

Introduction to Playbooks

We have ping servers so far.

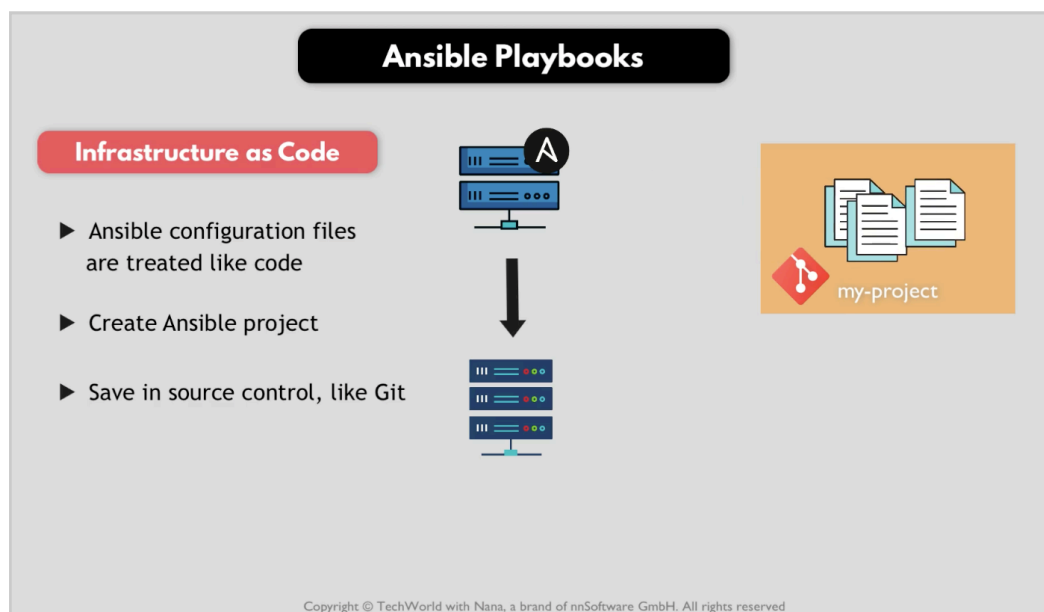
Next we want to configure the servers:

- Install applications
- Update
- Create files

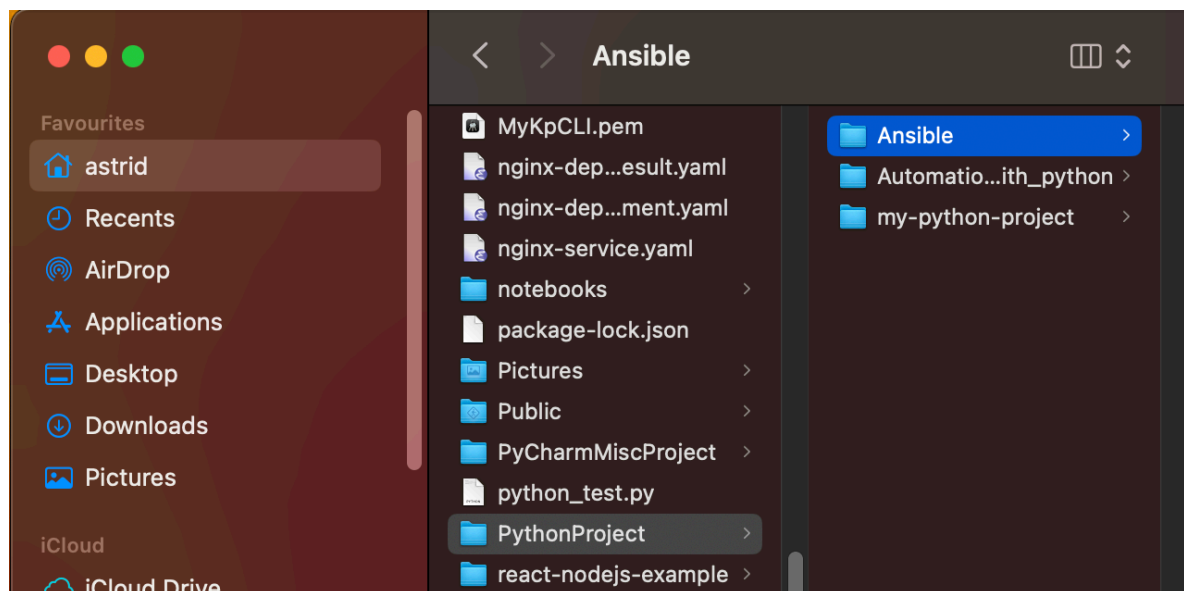
So we will write Ansible Playbooks

Ansible is IaC so we will treat Ansible files as code and we will create an Ansible folder for an Ansible project like we do when we create a java project.

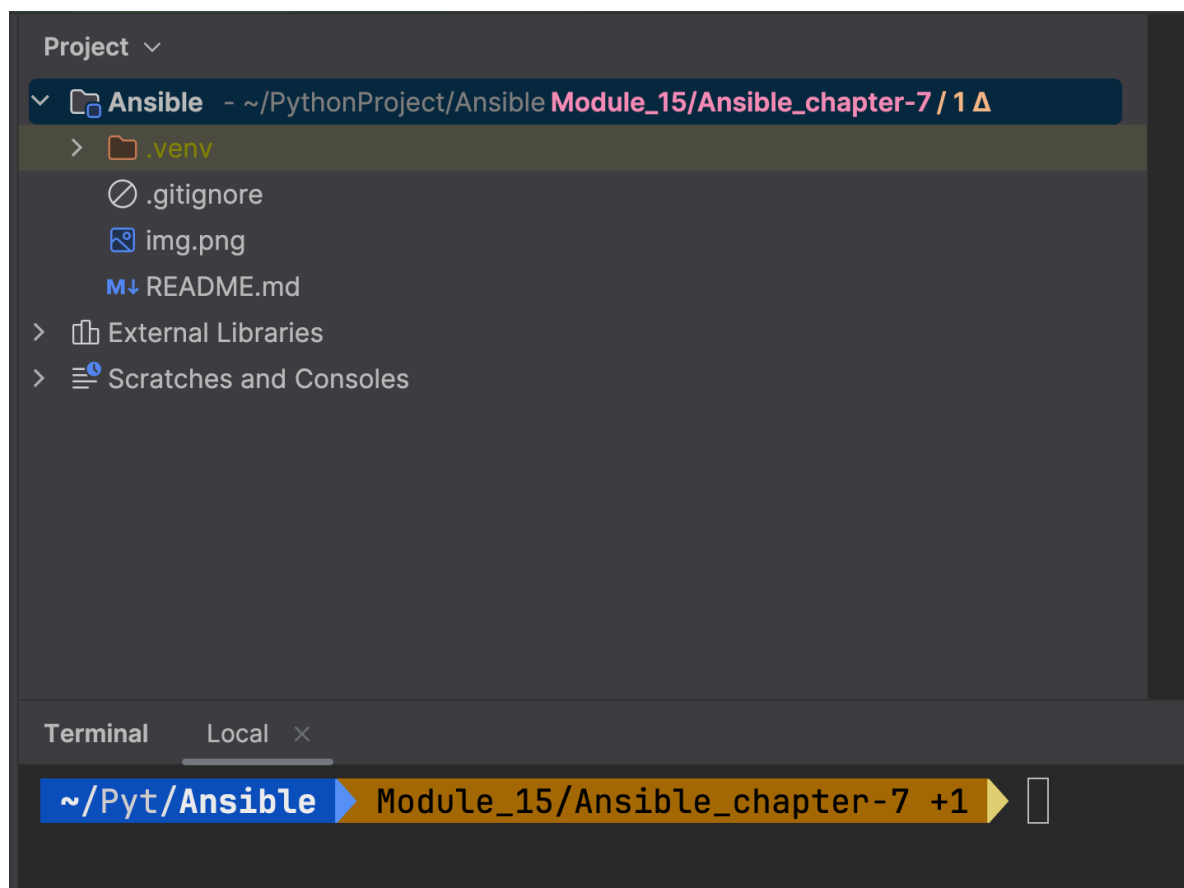
Then we can have source control for the project in Git



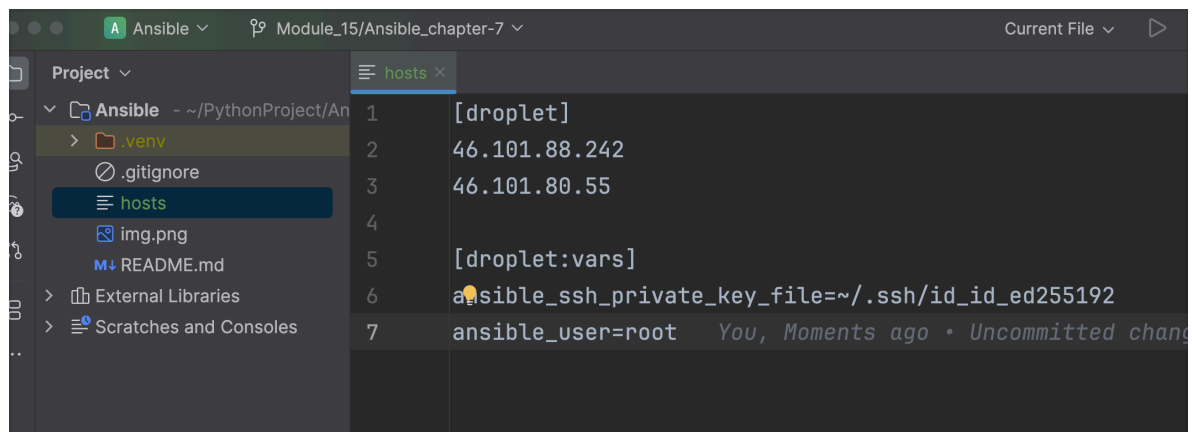
So I created a folder called 'Ansible' in /Users/astrid/PythonProject



And I create a branch for this chapter 7



Now I will create the 'hosts' file and copy the content of Module 15 - chapter 6 'hosts' file into my ansible project:

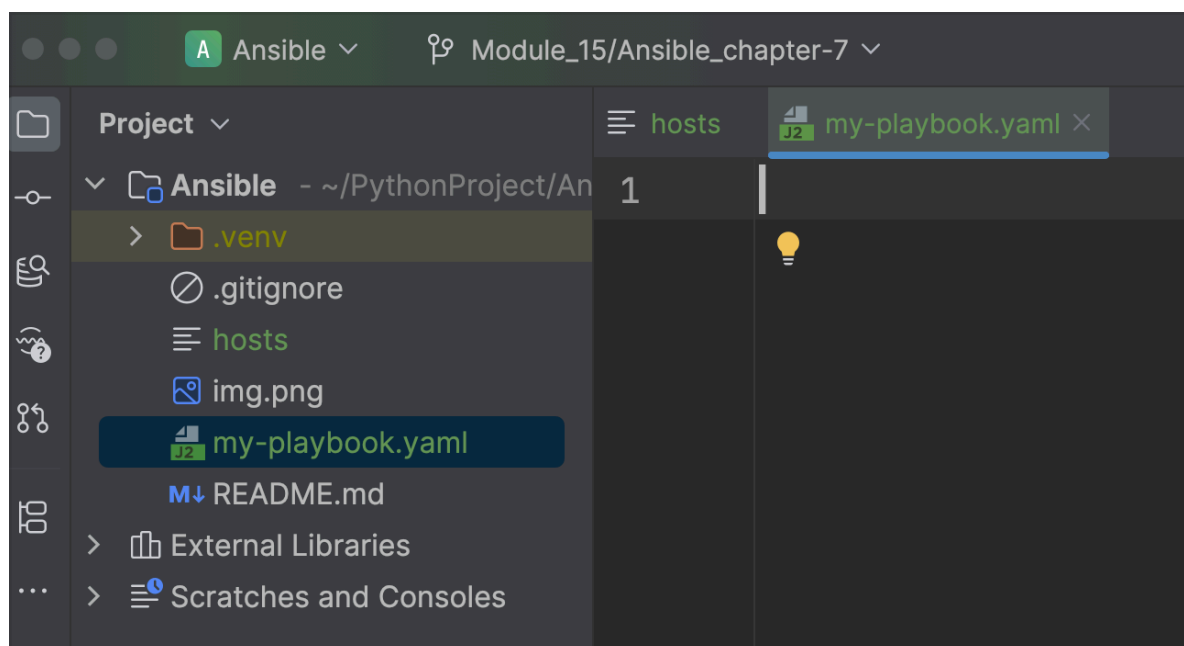


So we have the inventory -> hosts

Next we write the configuration, the tasks -> playbook

Create first playbook

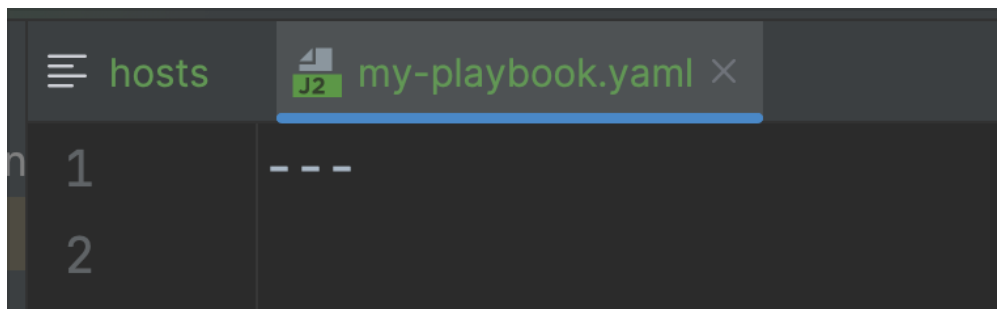
-> my-playbook.yaml



Playbook

- ▶ Ordered list of tasks
- ▶ Plays & tasks run in order from top to bottom
- ▶ Written in YAML

We use 3 dashes to separate blocks in YAML



Task -> unit

Group of tasks -> play

1 or more plays -> play book

Playbook

- ▶ A Playbook can have multiple Plays
- ▶ A Play is a group of tasks

For example we can have:

- 1 play for database servers where we install Postgres
- 1 play for web servers where we install nginx for example

1 Play for Database Servers

1 Play for Webservers

And both plays are in ! Playbook.

So let's write the first play

- It can have a name, it is optional but recommended
- It needs a hosts, which servers we want to use, we can use the group name of an individual IP address. which ever we use they must be already in the inventory 'hosts' file.
- Now we can define tasks -> list of commands that we want to execute on all the servers listed in the bullet above:
 - Yaml syntax for list is dash -
 - Every task has a name and it must be descriptive of what the command does

Tasks for Play

1) Install nginx

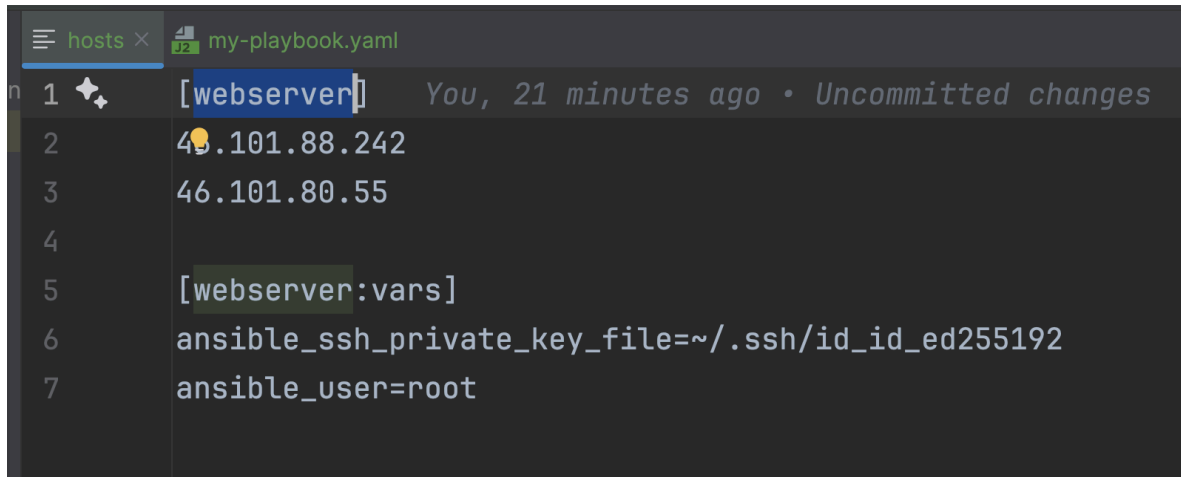
2) Start nginx

- We call the command which is a specific module
- If we install nginx manually we execute -> apt install nginx (ubuntu uses apt)
- In Ansible apt is a module so -> apt:
- we provide the name of the package we want to install ->

name:

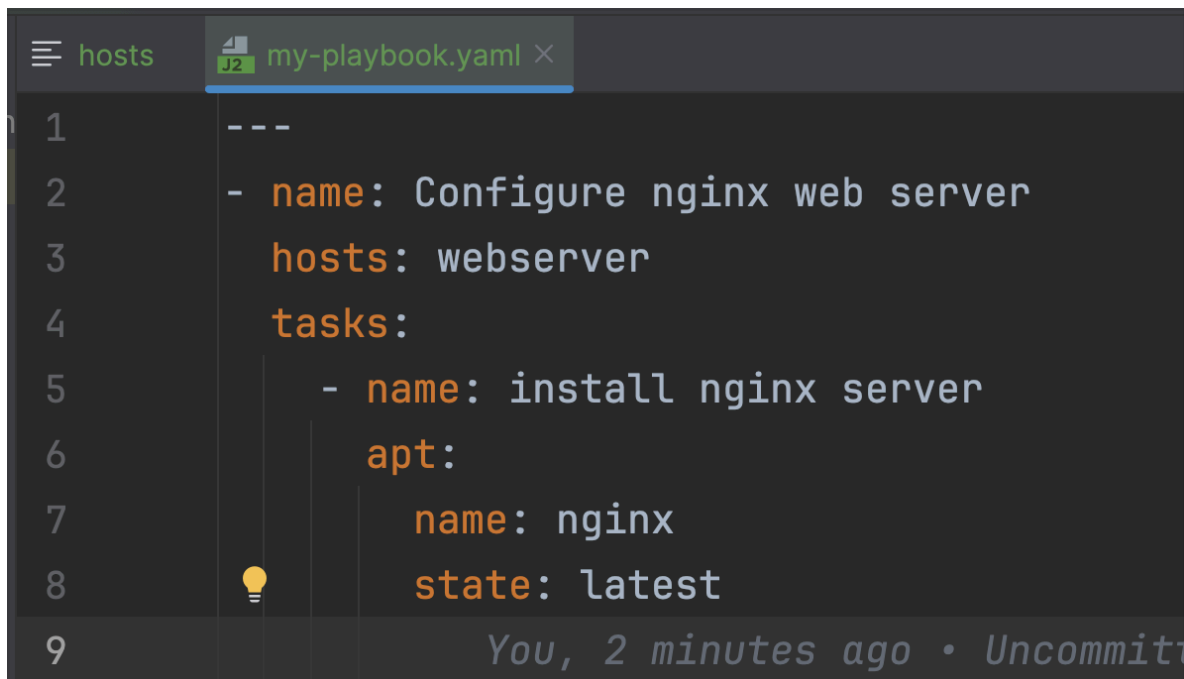
- we provide the state -> state: -> and we pass the value 'latest' to install the latest nginx package

So for the first task -> install nginx I rename the group from droplet to webserver



```
hosts x my-playbook.yaml
1 [webserver] You, 21 minutes ago • Uncommitted changes
2 45.101.88.242
3 46.101.80.55
4
5 [webserver:vars]
6 ansible_ssh_private_key_file=~/.ssh/id_id_ed255192
7 ansible_user=root
```

And this is the yaml code for the first task



```
hosts x my-playbook.yaml x
1 ---
2 - name: Configure nginx web server
3   hosts: webserver
4   tasks:
5     - name: install nginx server
6       apt:
7         name: nginx
8         state: latest
9 You, 2 minutes ago • Uncommitt
```

And for the second task -> start nginx

- And for this task the module is called 'service'
 - And it has a name: which is the name of the service
 - And it has state: which is the state we want the service to have in this case -> 'started'

```

9         - name: start nginx server
10           service:
11             name: nginx
12             state: started

```

Now we have our first playbook with 1 play and they play has 2 tasks

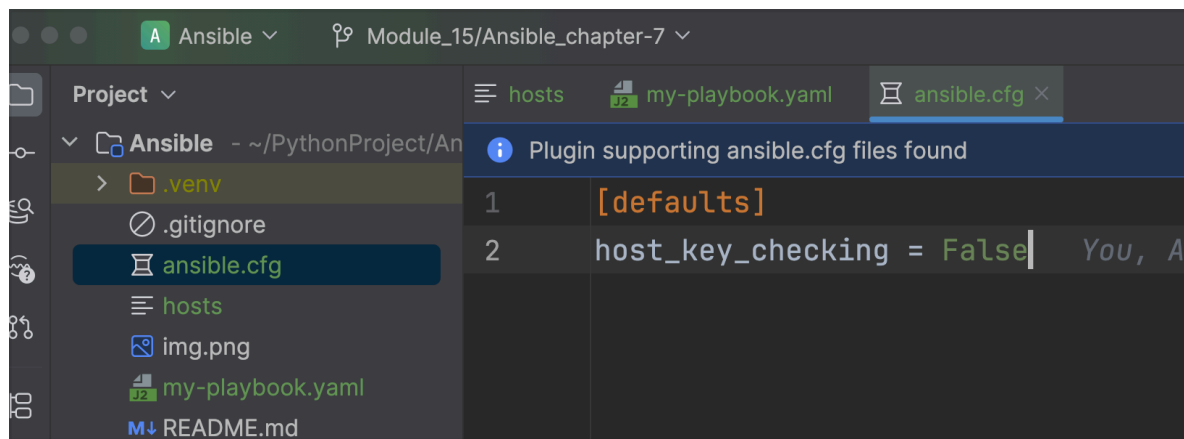
```

1 ---
2 - name: Configure nginx web server
3   hosts: webserver
4   tasks:
5     - name: install nginx server
6       apt:
7         name: nginx
8         state: latest
9     - name: start nginx server
10      service:
11        name: nginx
12        state: started

```

Execute the Playbook

Let's create our own ansible.cfg file in our project so each project has their specific configuration and we will copy the contents from Module 15 - chapter 6 ansible.cfg



And now we execute ansible playbook
-> `ansible-playbook -i hosts my-playbook.yaml`

```
Terminal Local x + v AI Agents v :  
$ ansible-playbook -i hosts my-playbook.yaml  
  
PLAY [Configure nginx web server] *****  
  
TASK [Gathering Facts] *****  
[WARNING]: Host '46.101.88.242' is using the discovered Python interpreter at '/usr/bin/python3.12', but future  
installation of another Python interpreter could cause a different interpreter to be discovered. See https://  
docs.ansible.com/ansible-core/2.21/reference\_appendices/interpreter\_discovery.html for more information.  
ok: [46.101.88.242]  
[WARNING]: Host '46.101.80.55' is using the discovered Python interpreter at '/usr/bin/python3.12', but future  
installation of another Python interpreter could cause a different interpreter to be discovered. See https://  
docs.ansible.com/ansible-core/2.21/reference\_appendices/interpreter\_discovery.html for more information.  
ok: [46.101.80.55]  
  
TASK [install nginx server] *****  
changed: [46.101.80.55]  
changed: [46.101.88.242]  
  
TASK [start nginx server] *****  
ok: [46.101.88.242]  
ok: [46.101.80.55]  
  
PLAY RECAP *****  
46.101.80.55      : ok=3   changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignor  
ed=0  
46.101.88.242    : ok=3   changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignor  
ed=0
```

We got the list of plays -> which is only ONE so far -> 'Configure nginx web server'

Then we get 'TASK [Gathering Facts]' where ansible gathers facts of the servers

"Gather Facts" Module

- ▶ Is automatically called by playbooks
- ▶ To gather useful variables about remote hosts that can be used in playbooks
- ▶ So Ansible provides many *facts* about the system, automatically

Then we have the first task 'TASK [install nginx server]'

And we see the servers IP addresses are now in yellow colour, meaning the server has changed. So installing nginx server was a change to a server.

Then the second task 'TASK [start nginx server]' -> shows the OK status

And finally we have a summary 'PLAY RECAP' where we can see the servers which are in yellow because there was at least one change and we can see that each server has changed=1.

Next, ssh into the servers to check:

-> ssh **root@46.101.80.55**

-> ps aux | grep nginx

ps aux lists all running processes on the system (see notes Module 2 - ps aux and netstat -lptn Q & A and Module 2 - OS - CPU - RAM - DISK - NETWORK Q&A)

- ps — process status
- a — show processes for all users (not just yours)
- u — display in user-oriented format (shows user, %CPU, %MEM, start time, etc.)
- x — include processes not attached to a terminal (daemons, background services)

Typical output columns: USER, PID, %CPU, %MEM, VSZ, RSS, TTY, STAT, START, TIME, COMMAND.

Common uses:

- Find a process's PID to kill it: `ps aux | grep node`
- Check what's eating CPU/memory
- Confirm a service is actually running

```
> ssh root@46.101.80.55
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 15 10:47:48 UTC 2026

System load:  0.0               Processes:    102
Usage of /:   4.4% of 47.39GB   Users logged in:  0
Memory usage: 15%              IPv4 address for eth0: 46.101.80.55
Swap usage:  0%                IPv4 address for eth0: 10.16.0.5

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Wed Jul 15 10:35:03 2026 from 140.228.47.252
root@ubuntu-s-1vcpu-2gb-lon1:~# ps aux | grep nginx
root      23835  0.0  0.3 11200 7100 ?        S    10:34   0:00 nginx: master process /usr/sbin/nginx -g daemon on; master_proc
ess on;
www-data  23838  0.0  0.2 12888 4428 ?        S    10:34   0:00 nginx: worker process
root      24073  0.0  0.1 7076 2164 pts/0    S+   10:48   0:00 grep --color=auto nginx
root@ubuntu-s-1vcpu-2gb-lon1:~#
```

And we can see that nginx process is running.

-> `nginx -v`

```
root@ubuntu-s-1vcpu-2gb-lon1:~# nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
```

And we can see the latest version that ansible installed on the server of nginx

Install specific version

For ubuntu we can check packages here: <https://launchpad.net/ubuntu>

← → ↺ 🏠 🌐 launchpad.net/ubuntu/+source/nginx

4 Channel 4 🎧 BBC iPlayer - Home 📅 Trello 🗒 Wise - Login 📖 Google NotebookL... 🔄 Overleaf - CV 🗣 Co



nginx package

[Overview](#)
[Code](#)
[Bugs](#)
[Blueprints](#)
[Translations](#)
[Answers](#)

nginx package in Ubuntu

libnginx-mod-http-geoip: GeoIP HTTP module for Nginx
 libnginx-mod-http-geoip-dbgSYM: debug symbols for libnginx-mod-http-geoip
 libnginx-mod-http-image-filter: HTTP image filter module for Nginx
 libnginx-mod-http-image-filter-dbgSYM: debug symbols for libnginx-mod-http-image-filter
 libnginx-mod-http-perl: Perl module for Nginx
 libnginx-mod-http-perl-dbgSYM: debug symbols for libnginx-mod-http-perl
 libnginx-mod-http-xslt-filter: XSLT Transformation module for Nginx
 libnginx-mod-http-xslt-filter-dbgSYM: debug symbols for libnginx-mod-http-xslt-filter
 libnginx-mod-mail: Mail module for Nginx
 libnginx-mod-mail-dbgSYM: debug symbols for libnginx-mod-mail
 libnginx-mod-stream: Stream module for Nginx
 libnginx-mod-stream-dbgSYM: debug symbols for libnginx-mod-stream
 libnginx-mod-stream-geoip: GeoIP Stream module for Nginx
 libnginx-mod-stream-geoip-dbgSYM: debug symbols for libnginx-mod-stream-geoip
 nginx: small, powerful, scalable web/proxy server
 nginx-common: small, powerful, scalable web/proxy server - common files
 nginx-core: nginx web/proxy server (standard version)
 nginx-dbgSYM: debug symbols for nginx
 nginx-dev: nginx web/proxy server - development headers
 nginx-doc: small, powerful, scalable web/proxy server - documentation
 nginx-extras: nginx web/proxy server (extended version)
 nginx-full: nginx web/proxy server (standard version with 3rd parties)
 nginx-light: nginx web/proxy server (basic version)

This package has [4 new bugs](#) and [0 open questions](#).

And here is the list of packages

Package information		
Maintainer:  Debian Nginx Maintainers	Urgency: Low Urgency	Upstream connections  Nginx → head High performance HTTP proxy server. Bug supervisor: ✓ Bug tracker: ✓ Branch: ✓ ✖ There are no registered releases for the Nginx → head. Show upstream links
Architectures: any all	Latest upload: 1.30.1-6ubuntu1	
actual publishing details may vary in this distribution, these are just the package defaults.		
Stonking Stingray (active development) Nginx head series		
↳  1.30.1-6ubuntu1	release (main)	2026-07-08
The Resolute Raccoon (current stable release) Nginx head series		
↳  1.28.3-2ubuntu1.6	security, updates (main)	2026-06-22
↳  1.28.3-2ubuntu1	release (main)	2026-04-09
The Questing Quokka (supported) Nginx head series		
↳  1.28.0-6ubuntu1.8	security, updates (main)	2026-06-22
↳  1.28.0-6ubuntu1	release (main)	2025-09-11
The Noble Numbat (supported) Nginx head series		
↳  1.24.0-2ubuntu1.13	security, updates (main)	2026-06-22
↳  1.24.0-2ubuntu1	release (main)	2024-04-04

So we can find the specific version and use it the playbook under the 'name' of the package

```

4      tasks:
5      - name: install nginx server
6        apt:
7      ✨      name: nginx=1.24.0-2ubuntu7.13

```

And we edit the state to 'present'

```

5      - name: install nginx server
6        apt:
7        name: nginx=1.24.0-2ubuntu7.13
8      ✨      state: present

```

Q:

I wonder if I will end up with 2 nginx with different versions when I re-run ansible playbook with different package version?

A:

No, you won't end up with two nginx versions — package managers don't work like that. There's only ever one version of nginx installed at a time under that package name.

Think of it like this:

1. You currently have nginx v1.24 installed.
2. You tell Ansible "install version 1.18 instead."
3. The package manager (apt/yum) swaps v1.24 out for v1.18 — same package, same files, just replaced. It's not "install a second copy alongside the first."
4. Along the way, it might briefly restart the nginx service (a few seconds of interruption), because installing the new files often triggers an automatic restart.

So the end state is simple: one nginx, now at the version you asked for, still running.

And I re-run

-> ansible-playbook -i hosts my-playbook.yaml

```

> ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

TASK [install nginx server] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

TASK [start nginx server] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

PLAY RECAP *****
46.101.80.55      : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignor
ed=0
46.101.88.242    : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignor
ed=0

```

Note: 'TASK [Gathering Facts]' don't show warnings any more -> I fix it in 'hosts' file by setting the python interpreter

```

hosts x my-playbook.yaml ansible.cfg
1 [webserver]
2 46.101.88.242
3 46.101.80.55
4
5 [webserver:vars]
6 ansible_ssh_private_key_file=~/.ssh/id_id_ed255192
7 ansible_user=root
8 ✨ ansible_python_interpreter=/usr/bin/python3.12 You, 45 minute

```

We also can see that all the servers are in green colour, no changes. That is because version 1.24.0-2ubuntu7.13 is the version with installed before with 'latest' so no changes really.

We can also use regular expression to define the version of the package

```

tasks:
5 - name: install nginx server
6   apt:
7     name: nginx=1.24.* You, 5 minute
8     state: present

```

And we re run the ansible playbook
-> ansible-playbook -i hosts my-playbook.yaml

```
hosts  my-playbook.yaml  ansible.cfg
2  - name: Configure nginx web server
6      apt:
7          name: nginx=1.24.*  You, 5 minutes ago • Uncommitted changes
8          state: present
9  - name: start nainx server

Terminal  Local  ×  +  v  AI Agents

) ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

TASK [install nginx server] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

TASK [start nginx server] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

PLAY RECAP *****
46.101.80.55      : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
ed=0
46.101.88.242    : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
ed=0
```

Ansible Idempotency

```
PLAY RECAP *****
46.101.80.55      : ok=3    changed=0    unreachable=0    failed=0
ed=0
46.101.88.242    : ok=3    changed=0    unreachable=0    failed=0
ed=0
```

We had zero changes, this happened because ansible detected that the version is already running in both servers so there is no change to be made.

Ansible didn't eve try to perform the action in this case didn't even try to install nginx version 1.24.* again

Idempotency

- ▶ **Most** Ansible modules check whether the desired state has already been achieved
- ▶ They exit without performing any actions

Idempotency



==



Actual State

Desired State

version 1.24 installed

version 1.24 installed



Nothing to do

Uninstalling

With 'apt' module we can also uninstall the packages. And also we can stop the service using the 'service' module.

To uninstall a package we use
State -> absent

And to stop the service we use
State -> stopped

```
4      tasks:
5          - name: uninstall nginx server
6            apt:
7              name: nginx=1.24.*
8              state: absent
9          - name: stop nginx server
10           service:
11             name: nginx
12             state: stopped    You, Moments ago
```

And we re-run it

```
> ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [46.101.88.242]
ok: [46.101.80.55]

TASK [uninstall nginx server] *****
changed: [46.101.88.242]
changed: [46.101.80.55]

TASK [stop nginx server] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

PLAY RECAP *****
46.101.80.55      : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    igno
ed=0
46.101.88.242    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    igno
ed=0
```

And there is one change -> nginx is uninstalled

And to check idempotency we re-run the same playbook and we will get the
same output -> current state is the same as the desired state


```
> ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

TASK [uninstall nginx server] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

TASK [stop nginx server] *****
ok: [46.101.80.55]
ok: [46.101.88.242]

PLAY RECAP *****
46.101.80.55      : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignor
ed=0
46.101.88.242    : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignor
ed=0
```